Scientific Principle

Based on established polymer-science principles, Nylon 66 exhibits increased molecular chain mobility when exposed to temperatures near or above its glass transition temperature (Tg). High-temperature exposure alters the thermal history of the material, increasing dye mobility and therefore raising the likelihood of color bleeding and migration. Our internal DOE observations align with this known behavior.

依據既有的聚合物材料科學原理, Nylon 66 在接近或高於其玻璃轉移溫度（Tg）時, 分子鏈活動度會顯著提高. 高溫會改變材料的熱歷史, 使染料更容易移動, 因而提高染色滲移與移染的可能性. 我們的 DOE 結果與此已知材料行為一致.

Vuori, Inc.	
Supplier Company Name: Vuori Inc.	
Sample Type: Bulk	
Division: Women's	
Department: Regular Apparel	**Product Category:** Sports Bras,Pants
Sample Description: One sample of knitted fabric BlissBlend Form™	
Country of Origin: Taiwan, China	**Country of Destination:** /
Season: FA25 Bulk fabric from Mas	**PO# / Ref#:** / **Bulk Lot#:** 2550640A
Style #: VW1370/VW4309/VW4414	**Style Name:** /
Color: Strawberry	**Article Number:** KP04570JA0
Agent Name: /	**Manufacturer:** YU YUANG TEXTILE CO.,LTD.
Construction: Interlock	**End Use:** Shell
Fabric Weight: 240gsm	**Fiber Content:** 74% Polyamide 26% Elastane
Fabric Width: /	**Cuttable Fabric Width:** 51"
Thickness of Leather: /	
Zipper Vendor Name: /	**Zipper Pull Vendor Name:** /
Submitted Functional Claims (if any): Moisture Wicking	

Findings & Observations:

Three DOE evaluations were conducted using BlissBlend Form 74% Polyamide (Nylon 66) + 26% Elastane dyed in Strawberry color. All specimens were immersed in approximately 70°C hot water for 15 minutes.

DOE 以鮮豔紅色之樣品進行三組內部實驗, 所有樣品皆於約 70°C 熱水中浸泡 15 分鐘.

1. Garment specimen exposed to two high-temperature container shipping cycles showed significant dye release into the water. 經歷兩次集裝箱高溫運輸之成衣, 出現明顯紅色染料釋放到水中.

2. Fabric specimen exposed to one high-temperature container shipping cycle showed less noticeable dye release. 經歷一次集裝箱高溫之布料, 亦可觀察到紅色染料釋放到水中.

3. Fabric specimen as keeping sample from Yu Yuang with no shipping exposure showed also very noticeable dye release: 未經集裝箱高溫, 裕源留樣之布料, 也一樣出現明顯紅色染料釋放到水中.

POV:

While the dry Tg of Nylon 66 is around 50 °C, hot-water conditions act as an amplifier that effectively reducing the Tg and increasing polymer chain mobility, therefore dye bleeding & migration observed at 70 °C is a normal & reasonable material behavior. 尼龍 66 的乾燥 Tg 約為 50°C, 但熱水條件像擴大器會降低有效 Tg 並提高分子鏈活動度, 因此在 70°C 觀察到染料釋出 & 遷移是正常 & 合理的材料行為.

The DOE results indicate that when specimens are immersed in approximately 70°C of hot water, the Strawberry colored dye released as color bleeding into water regardless the specimens have been exposed to high-temperature container shipping cycles or not . 實驗結果顯示, 當樣品浸入約 70°C 的熱水中後 無論樣品是否經歷過高溫貨櫃運輸循環, 紅色染料都會滲入水中.

That said, the possible root cause is not by the high temperature of the transportation, but the presence of removable dye or lack of washing/soaping/fixation in the material dyeing and finishing state itself, and the high temperature of the transportation may only increase the severity of failure. 顯示可能的根因並非由運輸的高溫造成, 而是材料/染整狀態本身已存在可移動染料或皂洗/固色不足, 運輸的高溫可能提升失效的嚴重度.

74% Nylon 66:

Falls within the wet glass-transition high-mobility region (approximately 50~70 °C), where polymer chain mobility increases and dye molecules become more mobile. 在 wet Tg 高活動區（50–70 °C）染料容易移動.

26% Elastane:

Does not form stable chemical bonds with acid dyes, but can physically absorb or retain unfixed dye. 吸附但不固定染料.

Under hot-water conditions, this physically retained dye is more easily released. 在熱水中更容易釋放.

Following observations occurred:

Self-bleeding (dye released into the water bath)

Occurrence even on non-shipped samples. 即使未經 shipment 的樣品也一樣發生.

The issue is more indicative of dye after-wash or fixation behavior rather than an inherent material abnormality. 更偏向 after-wash/soaping/dye fixation 的問題, 而非材料異常.

Fabric Composition as Nylon + Elastane blended × Acid dye in practical:

Nylon forms ionic bonds with acid dyes. Nylon 可與酸性染料形成離子鍵.

Elastane does not chemically fix acid dyes and may physically retain unfixed dye. Elastane 不會與酸性染料形成穩定化學鍵結, 但可能物理吸附未固著染料.

Under hot water conditions, this retained dye is more easily released. 在熱水條件下, 這些未固著染料更容易被釋放.

The observed self-bleeding behavior, including in non-shipped samples, indicates a dye fixation or after-wash efficiency issue rather than an inherent polymer defect. 即使未經運輸之留樣亦出現自體滲色, 顯示問題更偏向染料固著或後洗效率, 而非材料本質異常。

Focus!

After-wash/Soaping thoroughly **經過徹底的後洗／皂洗**

Eliminate weakly bound dyes **移除所有弱鍵結染料**

Prevent elastane "temporary dye" **避免彈性纖維產生 “暫時性著色” 現象**

Elastane is not designed to be dyed by conventional dye systems, industry’s best practice is to fully fix and wash off dyes on Nylon, so that Elastane does not act as a temporary dye keeper.

彈性纖維本身並不是為傳統染色系統設計的, 因此不能真正被酸性染料染色. 業界最佳做法是將尼龍上的染料完全固著並徹底清洗乾淨, 以避免彈性纖維暫時吸附未洗淨的浮色染料.

Below from the shared Excel Test Results – High Temperature: 70C is the main impactor caused the poor CF results VS. 6 major fibers:

Body Fabric	Style Number	Color	Lot#
			L/D standard
KP04570JA0-BlissBlend Form™	VW1370/ VW4309/ VW4414	Strawberry	2550640A from Mas
			25414491
			25414502
			2550563A
			2550641A
			2550460A

CF to Crocking		CF to Dye Transfer In Storage Option I: 24C for 48 hours (The wet specimen was placed in a plastic bag after applying pressure, then put into an oven. The temperature was raised to 24°C and maintained for 48 hours.)							CF to Dye Transfer In Storage Option I: 70C for 48 Hours (The wet specimen applying pressure and placed in a plastic bag, then put into an oven. The temperature was raised to 70°C and maintained for 48 hours.)						
Dry	Wet	Color Change	Acetate	Cotton	Nylon	Polyester	Acrylic	Wool	Color Change	Acetate	Cotton	Nylon	Polyester	Acrylic	Wool
4	4	4	4	4	4	4	4	4	N/A	N/A	N/A	N/A	N/A	N/A	N/A
4.5	3.5	4.5	4	4	3.5	4	3.5	4	4.5	3.5	3.5	3	4	3.5	3.5
4.5	3	4.5	4.5	4.5	4	4.5	4.5	4.5	4.5	4	3	1	3.5	3.5	2
4.5	3.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5	4	3	1.5	3.5	3.5	2
4.5	3.5	4.5	4	4	3.5	3.5	4	4	4.5	3	2	2	3	3	2
4.5	3.5	4.5	4	4	4	4	4	4	4.5	3	2	1.5	3	3	2
4.5	4	4.5	4	4	4	3.5	4	4	4.5	3	2.5	2.5	3	3	2.5